EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Insertion sort always swaps each element with its neighbour.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

# Insertion sort: trace and implement

Name: _____ Class: _____ Date: _____

**1. What is [5,2,4] after one complete ascending bubble-sort pass?**

**2. What is the value of i after i++ when i was 2?**

**3. Is a stack FIFO or LIFO?**

**4. What change makes bubble sort terminate early on sorted data?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

[2,4,5].

### 2. Model answer

3.

### 3. Model answer

LIFO: last in, first out.

### 4. Model answer

Track whether any swap occurred; if none occurred during a complete pass, stop.

### Vocabulary — know these terms

key — Value being inserted into the sorted prefix.

sorted prefix — Initial segment already placed in order.

shift — Move a value one index to create space.

in-place — Uses little additional data storage.

stable — Equal elements can preserve relative order.



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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why insertion sort is efficient on almost-sorted input.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Insert key 3 into [2,6,3,8], assuming [2,6] is sorted.

### Question 2

Trace insertion sort fully on [5,1,4].





NEW LEARNING

# Read, annotate and explain

### One key at a time

The prefix to the left of i is already sorted; take a[i] as the key.

### Make room

Shift larger values in the sorted prefix one position to the right.

### Insert the key

Place it at j+1; the sorted prefix grows by one element.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Insert key 1 into the sorted prefix [4,8] of [4,8,1,6].

### Predict first

Identify the shifts needed.

### Model solution

Shift 8 then 4 to the right and insert 1 at index 0 → [1,4,8,6].





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve